

WIRELESS NETWORKING

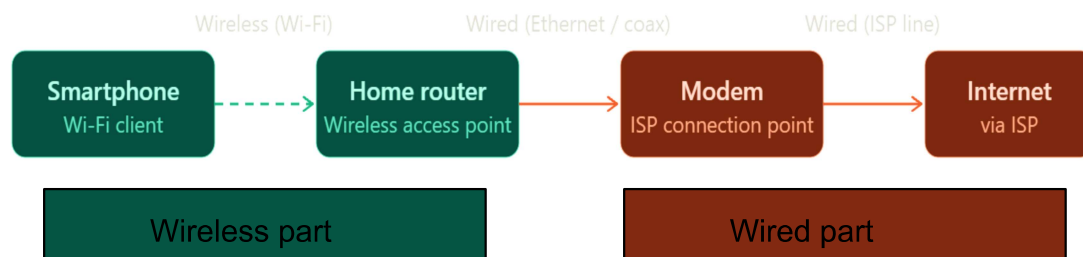
Task 1: List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

802.11n (Wi-Fi 4) operates on both the 2.4 GHz and 5 GHz bands, with a maximum theoretical speed of about 600 Mbps. In order to improve throughput and range over previous standards, it introduced MIMO (multiple antennas).

802.11ac (Wi-Fi 5): This technology only uses the 5 GHz band and has a maximum theoretical speed of roughly 3.5 Gbps (with wide channel bonding and multiple streams). Because of its higher-order modulation and wider channels (up to 160 MHz), it is substantially faster than 802.11n.

802.11ax (Wi-Fi 6) uses both 2.4 GHz and 5 GHz (Wi-Fi 6E extends into 6 GHz) and has a maximum theoretical speed of approximately 9.6 Gbps. It uses OFDMA (stands for **Orthogonal Frequency-Division Multiple Access**). It's a way of letting multiple devices share the same Wi-Fi channel *at the same time*, instead of taking turns, which is a significant improvement over 802.11ac. This is especially important for crowded home networks with numerous connected devices (phones, laptops, smart TVs, IoT sensors, etc.).

Task 2: Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



Task 3: Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Hint: Consider both signal strength and security when choosing.

open **Command Prompt** and run: `netsh wlan show networks mode=bssid`

It will provide a list of WIFI networks available along with signal strength, band, and encryption etc.

SSID	Signal Strength	Security Type
srikant	100%	WPA2-Personal
JioFiber-EQa5b	65%	WPA2-Personal
Tenda_0F2AB0	5%	WPA2-Personal
E90000220847335725	5%	WPA2-Personal

Airtel_adit_7947	35%	WPA2-Personal
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Signal strength—higher % means a stronger signal (e.g. 80–100% is excellent, 50–70% is okay but not great, and below 30% is often too weak for anything beyond basic browsing). A strong signal usually means faster internet and fewer drops or buffering.

Security type—from least to most secure:

- **Open / no password—anyone** can connect, and someone could potentially see your data. Only use this if it's a network you know and trust.
- **WPA2-Personal—a** good, widely used option. Safe enough for everyday home use.
- **WPA3-Personal—the** newest and strongest option. It's harder for someone to guess or crack the password compared to WPA2.

The final verdict: Security wins over a better signal from an open network. An open network means other people on it can potentially see what you're doing on the internet, which is dangerous for things like logging into accounts or banking apps.

So the simple rule is: “I would choose the network with WPA2 or WPA3 security, even if its signal is a bit weaker, because it keeps my data safe. “A fast but unprotected network is not worth the risk.” If both networks are well secured, opt for the one with the better signal, and if both WPA2 and WPA3 networks have similar signal strength, go for the WPA3 network as it's the newer and more secure option.

WEP (1997) → WPA (2003) → WPA2 (2004) → WPA3 (2018)

Task 4: Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID hiding

Every Wi-Fi router broadcasts its network name (the SSID) so nearby devices can see it in their list of available Wi-Fi networks and connect to it. SSID hiding is a router setting that stops this broadcast—the network still exists and still works, but it no longer shows up automatically in the list of "available networks" on phones or laptops.

Advantage

It keeps your network name out of plain sight for casual passersby. A neighbor or random stranger scanning for open Wi-Fi networks nearby won't see your SSID show up at all, so it's slightly less a target for someone to attempt to hack your network.

Disadvantage

It's really not a good for security point of view and adds real inconvenience. A hidden SSID doesn't actually stop a determined attacker—anyone using packet sniffing/capture tools can still detect that a hidden network exists by capturing the wireless traffic your own devices send while connecting.

Meanwhile, it makes life harder for legitimate use: every new device (a friend's phone, a smart TV, a new laptop) has to be manually configured with the exact SSID and password typed correctly, since it won't appear in the normal Wi-Fi list. Some devices also handle hidden networks poorly, causing slower or less reliable reconnections after Wi-Fi drops or the device restarts.